

Bias Framework: Definitions & Taxonomy

1. What Is Bias?

Bias refers to a preference or prejudice toward or against a person, group, or thing. Bias influences perceptions, judgments, and behaviors, often in ways that deviate from objective reasoning or equitable treatment.

Bias stems from:

- Societal stereotypes and cultural norms
- Personal experiences and upbringing
- Cognitive shortcuts (heuristics) that simplify complex information
- Institutional structures that perpetuate certain patterns

Key Insight from Capstone Research: Bias in communication is often indicated not by isolated words, but by sentiment, context, and repetition. Repeated negative sentiment toward a specific group or individual can constitute bias, even without explicit slurs.

2. Types of Bias

2.1 Explicit (Conscious) Bias

Definition: Prejudices and attitudes that a person is aware of and can articulate. These are processed at a conscious level and expressed through deliberate words or actions.

Characteristics:

- Consciously held beliefs
- Openly expressed preferences
- Intentional discriminatory behavior (e.g., hate speech, slurs, exclusion)

Examples:

- Using racial slurs
 - Stating that "women are less competent in leadership roles."
 - Refusing to hire someone based on their religion
-

2.2 Implicit (Unconscious) Bias

Definition: Attitudes or stereotypes that operate outside of conscious awareness. These biases can contradict a person's stated beliefs and values, yet still influence behavior and decision-making.

Characteristics:

- Operates automatically and unconsciously
- Shaped by background, culture, and media exposure
- Can manifest as microaggressions, subtle exclusion, or differential treatment
- Often revealed through reaction times, word associations, or pattern analysis rather than self-report

Examples:

- Assuming a woman in a meeting is an assistant rather than the lead engineer
- Unconsciously associating certain names with competence or criminality
- Preferring job candidates who attended the same university (affinity bias)

Capstone Application: Your research shows that sentiment analysis can reveal implicit bias patterns. For instance, consistently negative sentiment toward posts from certain demographic groups, even without explicit slurs, indicates implicit bias.

2.3 Institutional (Systemic) Bias

Definition: Discriminatory practices embedded in organizational policies, procedures, or cultural norms. These operate at the structural level and perpetuate disparities beyond individual prejudice.

Characteristics:

- Built into systems, not just individual actions
- May persist even when individuals are well-intentioned
- Creates cumulative disadvantage for certain groups

Examples:

- Promotion criteria that favor traits culturally associated with one gender
 - Recruiting only from networks that exclude certain racial groups
 - Performance review systems that penalize communication styles common in non-dominant cultures
-

3. Protected Characteristics (Groups Subject to Bias)

Bias typically manifests against individuals based on one or more of the following characteristics:

Category	Examples
Race & Ethnicity	Black, Asian, Hispanic, Indigenous, Middle Eastern, White, multiracial identity
Gender & Sex	Women, men, non-binary individuals, transgender individuals, cisgender individuals
Sexual Orientation	Lesbian, gay, bisexual, pansexual, asexual, heterosexual
Religion & Creed	Christianity, Islam, Judaism, Hinduism, Buddhism, atheism, agnosticism, other faiths
Disability	Physical disabilities (mobility, hearing, vision), mental health conditions, neurodivergence (autism, ADHD), chronic illnesses
Age	Younger workers, older workers, generational stereotypes
National Origin	Country of birth, ancestry, accent, immigration status
Genetic Information	Predisposition to certain health conditions

Proxy Attributes: Characteristics strongly correlated with protected attributes (e.g., zip code as a proxy for race; weightlifting ability as a proxy for gender) can also serve as vehicles for bias.

4. Manifestations of Bias in Text & Communication

4.1 Direct Bias Indicators

- Slurs and derogatory terms targeting protected groups
- Stereotyping language (e.g., "all X are Y")
- Erasure (excluding or ignoring certain groups entirely)

4.2 Indirect Bias Indicators

- Coded language that implies stereotypes without explicit terms (e.g., "urban," "articulate," "aggressive")
- Unequal standards (holding different groups to different expectations)
- Tone and sentiment patterns: Consistently negative or dismissive tone when referring to specific groups

4.3 Contextual & Behavioral Patterns

- Polarization language: Framing groups as adversaries ("us vs. them")
 - Piling on: Multiple negative interactions directed at one person or group (metadata signal of toxicity)
 - Frequency effects: Repeated exposure to biased language, even if mild per instance, creates cumulative harm
-

5. Measuring & Detecting Bias

5.1 Self-Report (Explicit Bias)

Surveys and questionnaires where individuals directly state their attitudes.

Limitations: Social desirability bias; people may not report prejudices they consciously hold.

5.2 Implicit Association Tests (Implicit Bias)

Reaction-time measures that reveal unconscious associations (e.g., IAT).

Limitations: Test-retest reliability concerns; scores may not predict real-world behavior.

5.3 Computational Text Analysis (Both Types)

Natural language processing, sentiment analysis, and LLM-based extraction to identify bias patterns in text.

Strengths:

- Can analyze large volumes of text
- Detects sentiment and contextual patterns beyond keyword matching
- Grounds findings in specific text spans

Limitations:

- Models may inherit biases from training data
- Requires human review for nuanced cases
- Sentiment subjectivity

Capstone Insight: Sentiment is a crucial determinant in predicting hate speech and toxicity. Metadata signals (e.g., number of toxic replies) can indicate behavioral bias patterns.

6. Language Examples by Protected Group

Race & Ethnicity

- Biased: "Thug," "inner city," "illegal" (as noun), "articulate" (when implying surprise)
- Context-dependent: "Urban," "minority" (may be appropriate in demographic analysis, problematic in coded usage)

Gender & Sex

- Biased: "Bossy" (for assertive women), "guys" (for mixed groups), "hysterical," "shrill."
- Gendered roles: "Chairman," "manpower," "man the desk."

Sexual Orientation

- Biased: Using "gay" as a pejorative, "lifestyle choice," assuming heterosexuality as the default

Religion

- Biased: "Fundamentalist" (used exclusively for one religion), religious holidays treated as universal defaults

Disability

- Biased (ableist): "Crazy," "insane," "lame," "confined to a wheelchair," "suffers from [condition]."
- Medical metaphors: "Blind to the facts," "tone-deaf," "falling on deaf ears."

Age

- Biased: "Too old to learn," "millennials are lazy," "OK Boomer" (dismissive)
-

7. Sentiment & Toxicity Framework (From Capstone Research)

Toxicity: Communication that is rude, disrespectful, or likely to make someone leave a discussion.

Key Finding: High-frequency exposure to hate speech correlates with increased PTSD symptomology.

Detection Method:

- Sentiment analysis identifies a negative tone
 - Metadata signals (e.g., number of toxic parent tweets or replies) predict toxicity levels
 - Context matters: Individual borderline comments may not be toxic, but repeated negative interactions constitute toxic behavior
-

8. Summary: Bias Detection Criteria

When analyzing text for bias, consider:

1. Explicit content: Are there slurs, stereotypes, or derogatory terms?
2. Implicit sentiment: Is the tone consistently negative toward a protected group?
3. Context & frequency: Are there patterns of exclusion, dismissiveness, or "piling on"?
4. Protected characteristics: Which groups are mentioned or implied?
5. Impact potential: Could this language create polarization, alienation, or psychological harm?